



HWA CHONG INSTITUTION
Preliminary Examination
Higher 1

CANDIDATE NAME

CT GROUP

11S

CENTRE NUMBER

INDEX NUMBER

CHEMISTRY

8872/02

Paper 2

11 September 2012

2 hours

Candidates answer **Section A** on the Question Paper.

Additional Materials: Data Booklet

Writing paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT group on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue, correction fluid or tapes.

Section A

Answer **all** questions.

Section B

Answer **two** questions on separate answer paper.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINERS' USE ONLY

Paper 1	Paper 2		TOTAL
Multiple Choice	Section A (Structured)	Section B (Free Response)	110
	Q1 / 11	Q5 / 20	
	Q2 / 12	Q6 / 20	
	Q3 / 6	Q7 / 20	
	Q4 / 11		
/ 30	Subtotal / 40	Subtotal / 40	

Section A

Answer **all** the questions in this section in the spaces provided.

- 1** The element iron is a transition metal. A number of naturally occurring isotopes have been identified as listed below.

Isotope	relative isotopic mass	natural abundance (%)
^{54}Fe	53.94	5.85
^{56}Fe	55.93	91.75
^{57}Fe	56.94	2.12
^{58}Fe	57.93	0.28

- (a)** Use the data above to calculate the relative atomic mass of iron to **two** decimal places.

[2]

- (b)** Describe the bonding in the element iron. Draw a diagram to illustrate your answer.

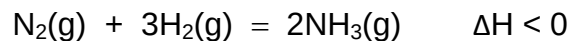
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[2]

Iron is used as a catalyst in the Haber-Bosch process for the industrial production of ammonia as shown below.



- (c) (i) Explain in detail, with the aid of an energy distribution diagram, why a catalyst is used in such an industrial process.

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- (ii) Le Chatelier's principle predicts that the highest equilibrium yield in the Haber-Bosch process should occur at a high pressure and at a low temperature. In practice, however, these conditions are not used. Explain why.

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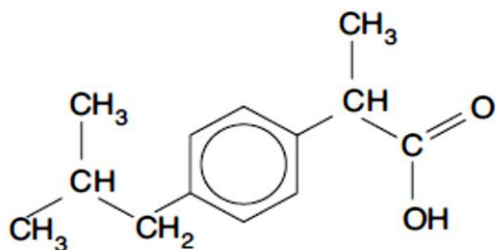
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[7]

[Total: 11]

- 2 Ibuprofen is one of the most commonly used non-steroidal anti-inflammatory drugs, used to treat chronic arthritic pain caused by inflammation of the joints.



ibuprofen

- (a) (i) Write down the molecular formula of ibuprofen.

.....

- (ii) Calculate the M_r of ibuprofen and use it to calculate how many grams are needed to make 100 cm^3 of a 0.15 mol dm^{-3} solution.

- (iii) Vigorous oxidation of ibuprofen produces a dibasic acid **A**. A solution containing 0.10 g of **A** required 12.0 cm^3 of 0.10 mol dm^{-3} NaOH for neutralisation.

Suggest a structure for **A**, showing your working.

[6]

(b) The K_a value for ibuprofen is $6.3 \times 10^{-6} \text{ mol dm}^{-3}$.

(i) Write an expression for K_a .

(ii) The pH of a 0.15 mol dm^{-3} solution of ibuprofen is 3.0, whereas the pH of a 0.15 mol dm^{-3} solution of HCl is 0.82.

Calculate the ratio of hydrogen ion concentrations in these two solutions and explain the significance of the ratio.

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[3]

(c) To avoid problems with digestive irritation over a long period of use, research is being carried out into ways of administering ibuprofen using skin patches. For this use the compound is dissolved in a hydrophilic gel which acts as a buffer.

(i) What do you understand by the term *buffer*?

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The buffer used in the pharmaceutical preparation is a solution containing Na_2HPO_4 and NaH_2PO_4 . These salts contain the HPO_4^{2-} and H_2PO_4^- ions respectively.

(ii) Write equations to show how this buffer reacts with

I. H^+ ions:

II. OH^- ions:

[3]

[Total: 12]

- 3** An unsaturated hydrocarbon **Z** is obtained by cracking hexane and is important in the chemical industry.

The standard enthalpy change of combustion of **Z** is $-2059 \text{ kJ mol}^{-1}$.

- (a)** Define the term *standard enthalpy change of combustion*.

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[2]

- (b)** When 0.47 g of **Z** was completely burnt in air, the heat produced raised the temperature of 200 g of water by 27.5°C .

- (i)** Calculate the amount of heat released in this experiment.

- (ii)** Use the data above and your answer to **(i)** to calculate the relative molecular mass of **Z** and hence give its molecular formula.

[4]

[Total: 6]

- 4 (a) An alcoholic beverage is a drink containing ethanol, commonly known as alcohol. Alcoholic beverages are divided into three general classes: beers, wines, and spirits.

The typical alcohol content in various alcoholic beverages are listed below:

alcoholic beverage	percentage by volume
Wines	16%
Beer	7%
Spirits	20%

If a bottle of wine is sold as "contains 16% alcohol", this means that there are 16g of ethanol per 100 cm³ of wine. A typical bottle contains 725 cm³ of wine while a typical can of beer contains 330 cm³ of beer.

- (i) Calculate how many grams of ethanol would be present in a typical bottle of wine which is said to contain 16% alcohol.
- (ii) Calculate the number of moles of ethanol in the bottle in (a)(i).
- (iii) Calculate the number of cans of beer that would contain the same amount of ethanol as in a bottle of wine.

[4]

- (b)** The amount of ethanol present in an alcoholic beverage can be investigated in a school laboratory using the redox titration method.

A student titrated a 25 cm^3 sample of a particular bottle of spirit with 1.50 mol dm^{-3} acidified potassium dichromate(VI) solution.

The oxidation of ethanol to ethanoic acid is given below.



- (i)** Write the half-equation for the reduction of dichromate(VI), $\text{Cr}_2\text{O}_7^{2-}$, ions.

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- (ii)** Construct the balanced equation for the redox reaction.

- (iii)** What is the volume, in cm^3 , of 1.50 mol dm^{-3} acidified potassium dichromate(VI) solution needed to convert all the ethanol present into ethanoic acid?

[4]

- (c) Ethanol can be oxidised successively to ethanal and then to ethanoic acid. The boiling points of these compounds are shown below.

compound	boiling point/ °C
ethanol	78
ethanal	21
ethanoic acid	118

Explain in terms of the intermolecular forces why the order of the boiling point is ethanal < ethanol < ethanoic acid.

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[3]

[Total: 11]

Section B

Answer **two** questions from this section on separate answer paper.

- 5 (a) (i) Using the chlorides of sodium, aluminium and silicon as examples, describe the reactions of the chlorides of the third period of the Periodic Table with water. Write equations where appropriate.
- (ii) Suggest what influence the type of bonding present in these chlorides has on their reaction with water.

[6]

- (b) Draw a dot-and-cross diagram to show the bonding in nitrogen trichloride, NCl_3 . State and explain the shape of this molecule and the bond angle it contains.

[2]

- (c) Compound **D** has molecular formula $\text{C}_4\text{H}_7\text{Cl}$ and decolourises aqueous bromine. On warming with acidified potassium manganate(VII), the purple solution turns colourless while a colourless and odourless gas which forms a white precipitate with limewater is evolved.

When compound **D** is warmed with aqueous sodium hydroxide, compound **E** is formed. When treated with alkaline aqueous iodine, compound **E** gives a pale yellow precipitate.

Oxidation of **E** gives substance **F**, which cannot be oxidised further, by warming with dilute acidic potassium dichromate(VI) but which forms an orange precipitate, **H**, with 2,4-dinitrophenylhydrazine.

When compound **D** is warmed with potassium cyanide in ethanol, compound **G**, which forms a carboxylic acid when warmed with dilute sulfuric acid, is formed.

Identify and suggest structures for **D**, **E**, **F**, **G** and **H**. Show how you deduced these structures, write equations for all of the reactions described above and suggest the types of reactions that are occurring.

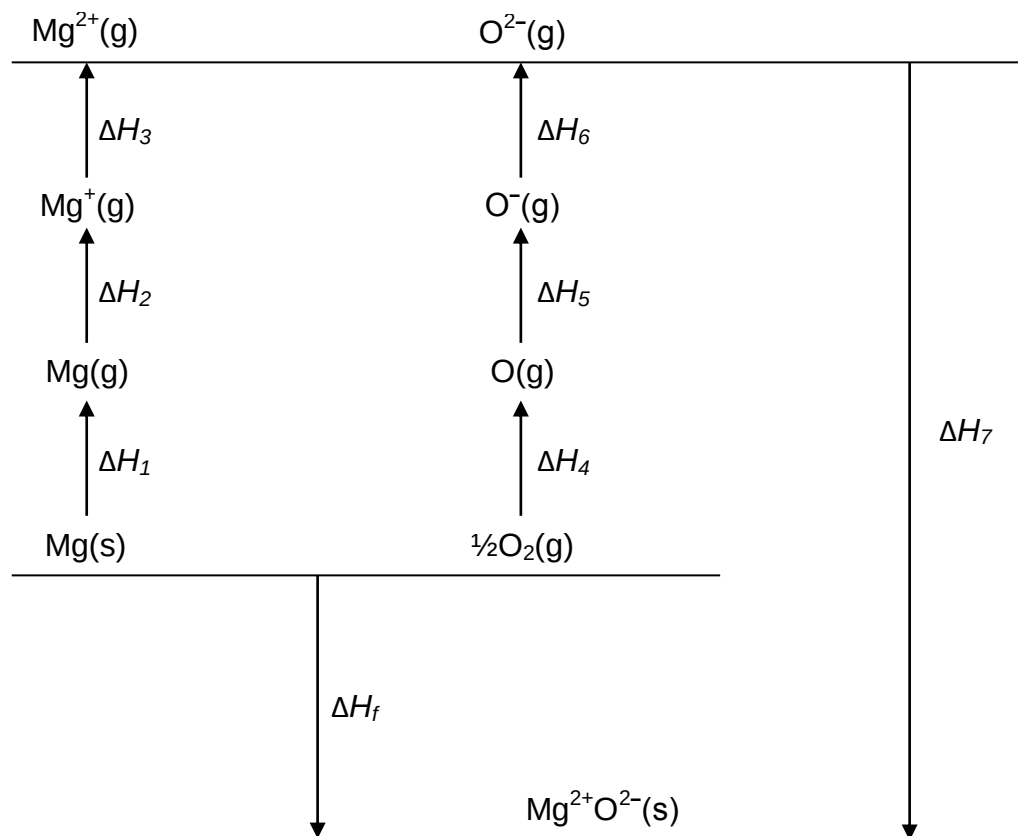
[12]

[Total: 20]

- 6 (a) Using the oxides of sodium, magnesium and sulfur as examples, describe the reactions of the oxides of elements in the third period of the Periodic Table with water. Write equations where appropriate.

[4]

- (b) The *enthalpy change of formation of magnesium oxide* may be represented by a Born-Haber energy cycle as described below.



- (i) Define the *enthalpy change of formation of magnesium oxide*.
- (ii) Name the enthalpy changes represented as ΔH_1 , ΔH_2 and ΔH_7 .
- (iii) Using relevant data from the Data Booklet, calculate ΔH_4 .
- (iv) Use ionisation energy values from the Data Booklet together with the values
 $\Delta H_1 = +150 \text{ kJ mol}^{-1}$
 $\Delta H_7 = -3889 \text{ kJ mol}^{-1}$
 $\Delta H_4 + \Delta H_5 + \Delta H_6 = +950 \text{ kJ mol}^{-1}$
 to calculate the enthalpy change of formation, ΔH_f , of magnesium oxide.
- (v) The value of ΔH_7 of barium oxide is $-3152 \text{ kJ mol}^{-1}$. Account for the difference in the values of ΔH_7 for magnesium oxide and barium oxide.

[9]

[Turn over]

- (c) Halogenoalkanes can react with aqueous silver nitrate to give precipitates. The rate of formation of the precipitates are given below:

compound	observation after adding $\text{AgNO}_3(\text{aq})$
$\text{CH}_3\text{CH}_2\text{I}$	ppt forms almost immediately
$\text{CH}_3\text{CH}_2\text{Br}$	ppt forms after 2 minutes
$\text{CH}_3\text{CH}_2\text{Cl}$	ppt forms after 10 minutes

Using suitable data from the Data Booklet, explain the observations as fully as you can.

[3]

- (d) (i) Suggest reagents, and describe the reaction conditions necessary, to convert 2-bromopentane into an alkene.
- (ii) Give the structural formulae of all the possible alkenes formed when 2-bromopentane undergoes the reaction in (d)(i).

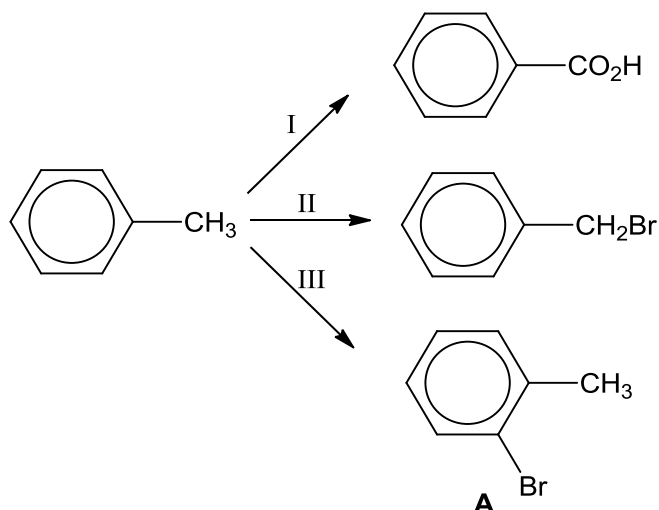
[4]

[Total: 20]

- 7 (a) Describe and explain the trends in atomic radius **and** first ionisation energy across period three of the Periodic Table.

[6]

- (b) Methylbenzene can be used as an additive in unleaded petrol. Some chemical transformations of methylbenzene are given below.



- (i) Suggest reagents and conditions for each of the reactions I, II and III, and state the type of reaction undergone in each case.
- (ii) There are two other positional isomers of **A**. Draw their structural formulae, and suggest which one is more likely to be formed along with **A** in reaction III.

The progress of reaction III can be followed by plotting the concentration of methylbenzene against time.

Data from the experiment are shown in the table.

Time/min	8	11	13	16	21	25
[methylbenzene]/ $\times 10^{-3} \text{ mol dm}^{-3}$	65	55	50	38	25	10

- (iii) Plot a graph of these results.
- (iv) Explain what is meant by the term *order of* reaction. Use your graph to deduce the order of reaction for methylbenzene.
- (v) Describe and explain what happens to the shape of the graph if the initial concentration of methylbenzene was doubled.

[14]

[Total: 20]